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1. The centroid centering trick is used to:

1 / 1 point

- ○ Find the rotation matrix for 3D to 2D projection.
- ◉ Translate 3D points so that the centroid becomes the origin.
- ○ Calculate the average distance between 3D points.
- ○ Project 3D points onto a 2D plane.

Correct
B) is the correct explanation. The primary purpose of the centroid centering trick is to translate (shift) all 3D points in a dataset so that the centroid of those points becomes the origin (0, 0, 0). This translation simplifies subsequent calculations and is especially useful in various computer vision and graphics tasks.

2. You have a set of 3D points:

1 / 1 point

Point 1: [5, 3, 2]

Point 2: [7, 4, 1]

Point 3: [6, 5, 3]

Apply the centroid centering trick and then calculate the new coordinates of point 3 after centering.

- ○ [1, 1, 1]
- ○ [1, 1, 0]
- ○ [1, 0, 1]
- ◉ [0, 1, 1]

Correct
To find the new position after re-centering:

New Coordinate = Original Coordinate - Centroid Coordinate

You can calculate the centroid of N points, we take the average of positions in respective dimensions as:

Centroid (Xc, Yc, Zc) = (ΣX, ΣY, ΣZ) / N